

9-HaplotypeCaller

Step Description

- **Role and Importance:** The step of HaplotypeCaller is the main variant calling step for our variable-depth low-cov WGS target regions (exome). It results in gvcf files that are considered intermediates for the final calling.
- **Tools or algorithms:** GATK HaplotypeCaller for variant calling.

Source Code & Workflow

- **GitLab Main Repository:** haplotypcaller-s3-mount
- **GitLab Branch:** main
- **GitLab Directory:** haplotypcaller-s3-mount. [See Page In Gitlab](#)
- **Argo Reference Workflow:** haplotypcaller-s3-mounted. [See Template in Argo](#)
- **Argo Master Workflow:** haplotypcaller-s3-mounted-multiple-tubes. [See Template in Argo](#)
- **EXAMPLE**

Inputs & Outputs

Overall volume

680-750 tubes (Good samples that are used in imputation)

Input:

- **Markdup BAM:**
 - **name:** {tubeid}. {batchid}. markdup.hg19.bam{.bai}
 - **Format:** BAM
 - **Avg size:** 1.5-3 GB
 - **Destination:** pipeline-v3.0/{batchid}/{tubeid}/fq2bam/markdup/full_bam/

Output:

- **GVCF:**
 - **name:** {tubeid}. {batchid}. hg19.gvcf.gz{.tbi}
 - **Format:** GVCF.gz
 - **Avg size:** 2 MB >
 - **Destination:** pipeline-v3.0/{batchid}/{tubeid}/gvcf/

Execution Environment

Workflow Orchestration System:

- **Cluster:** New Argo/K8
- **Node group:** fq2bam
- **Resource usage (per pod):** 5 CPU / 20 RAM (in fact, it's CPU * 4)

Environmental variables (Input parameters):

- **SAMPLE_IDS:** Json with delays. See [Delays](#).
- **BATCH_ID:** batchid that was assigned. It takes this format: batchddmmyy (ex. batch260625)
- **INPUT_DIR_S3:** path for BAM. Default is: s3://pipeline-v3.0/{batchid}
- **OUTPUT_DIR_S3:** path for gvcf. Default is: s3://pipeline-v3.0/{batchid}
- **PLOIDY_DIR_S3:** path for ploidy for imputation report, to determine ploidy num for sample. Default is: s3://pipeline-v3.0/batch_QC
- **ASSEMBLY:** genomic assembly to align by: hg19 or hg38 (default for now is hg19)
- **RECALIBRATED:** (yes/no) whether it's markdup only (default) or also recalibrated. Default is "no"
- **CPUS:** Number of CPUS to use for pod
- **BAM_SUFFIX:** the suffix that is assigned when we have markdup. Default is "markdup"
- **DBSNP:** (keep default) name of dbsnp vcf used for annotation inside this dir: s3://pipeline-v3.0/db/{ASSEMBLY}/.
- **REF:** (This is redundant) reference fasta
- **delay:** just a filler redundant parameter. (ignore)

Performance Metrics:

- **Per single unit:** ~5-10 min

- **Per full batch:** ~30-45 min

Additional Notes

- **Dependencies:** fq2bam and determining good samples.
- **Known edge cases or failure causes:** The main limitation to be appointed to this container is the ploidy issue. This container was not equipped with a mode to handle chrX in males, resulting in all called files becoming diploid for both male and females, regarding chrX. Performance issues were not detected lately.